Advanced Computer-Aided VLSI System Design
Lab4: Innovus Lab (1/2)

TA: Chia-Ying Huang r13943003@ntu.edu.tw

Data Preparation

1. Extract data from the folder **Innovus_Lab**.
2. The extracted directory contains
   - **design**
     A. CHIP_syn.v
     B. CHIP.io
     C. CHIP_func.sdc
     D. CHIP_scan.sdc
     E. CHIP-scan.def
   - **library**
     - **lef**
       A. N16ADFP_APR_Innovus_11M.10a.tlef
       B. N16ADFP_BondPad.lef
       C. N16ADFP_StdCell.lef
       D. N16ADFP_StdIO.lef
     - **lib**
       A. N16ADFP_StdCellff0p88v125c_ccs.lib
       B. N16ADFP_StdCellff0p88vm40c_ccs.lib
       C. N16ADFP_StdCellss0p72v125c_ccs.lib
       D. N16ADFP_StdCellss0p72vm40c_ccs.lib
       E. N16ADFP_StdIOff0p88v1p98v125c.lib
       F. N16ADFP_StdIOff0p88v1p98vm40c.lib
       G. N16ADFP_StdIOss0p72v1p62v125c.lib
       H. N16ADFP_StdIOss0p72v1p62vm40c.lib
     - **memory**
       A. rf_2p_hse.lef
       B. rf_2p_hse_nldm_ff_0p88v_0p88v_125c_syn.lib
       C. rf_2p_hse_nldm_ff_0p88v_0p88v_m40c_syn.lib
       D. rf_2p_hse_nldm_ss_0p72v_0p72v_125c_syn.lib
       E. rf_2p_hse_nldm_ss_0p72v_0p72v_m40c_syn.lib
       F. rf_2p_hse.gds
   - **sim**
     - **pre_sim**
       A. CHIP.tcf

- **post_sim**
  A. a.map
  B. CHIP_sim.v
  C. golden_result
  D. parameter.v
  E. vlog.f
  F. xrun.tcl
- mmmc.view

Introduction

In this lab, you will learn how to use Innovus to run the APR flow in N16 ADFP, and generate the required data for demonstration.

Data Preparation (Library)

1. Copy libraries to the folder **Innovus_Lab**.
   1.1 % **unzip Innovus_Lab.zip**
   1.2 % **cd Innovus_Lab**
2. Start Innovus: (change the directory to **Innovus_Lab**)
   2.1 % **innovus** (remember do not use background execution)
   2.2 Fail to open Innovus:
   % **cb**
3. Design Import
   3.1 ***File → Import Design…***
   3.2 Verilog
   - Files: **design/CHIP_syn.v**
   - Top Cell: ◆ By User: **CHIP**

   3.3 Technology/Physical Libraries
   - LEF Files:

/share1/tech/ADFP/supplementary/modified_lef/N16ADFP_APR_Innovus_11M.10a.tlef (first)

/share1/tech/ADFP/supplementary/modified_lef/N16ADFP_StdCell.lef

/share1/tech/ADFP/supplementary/modified_lef/N16ADFP_StdIO.lef

/share1/tech/ADFP/supplementary/modified_lef/N16ADFP_BondPad.lef

/share1/tech/ADFP/supplementary/modified_lef/rf_2p_hse.lef

3.4 Power

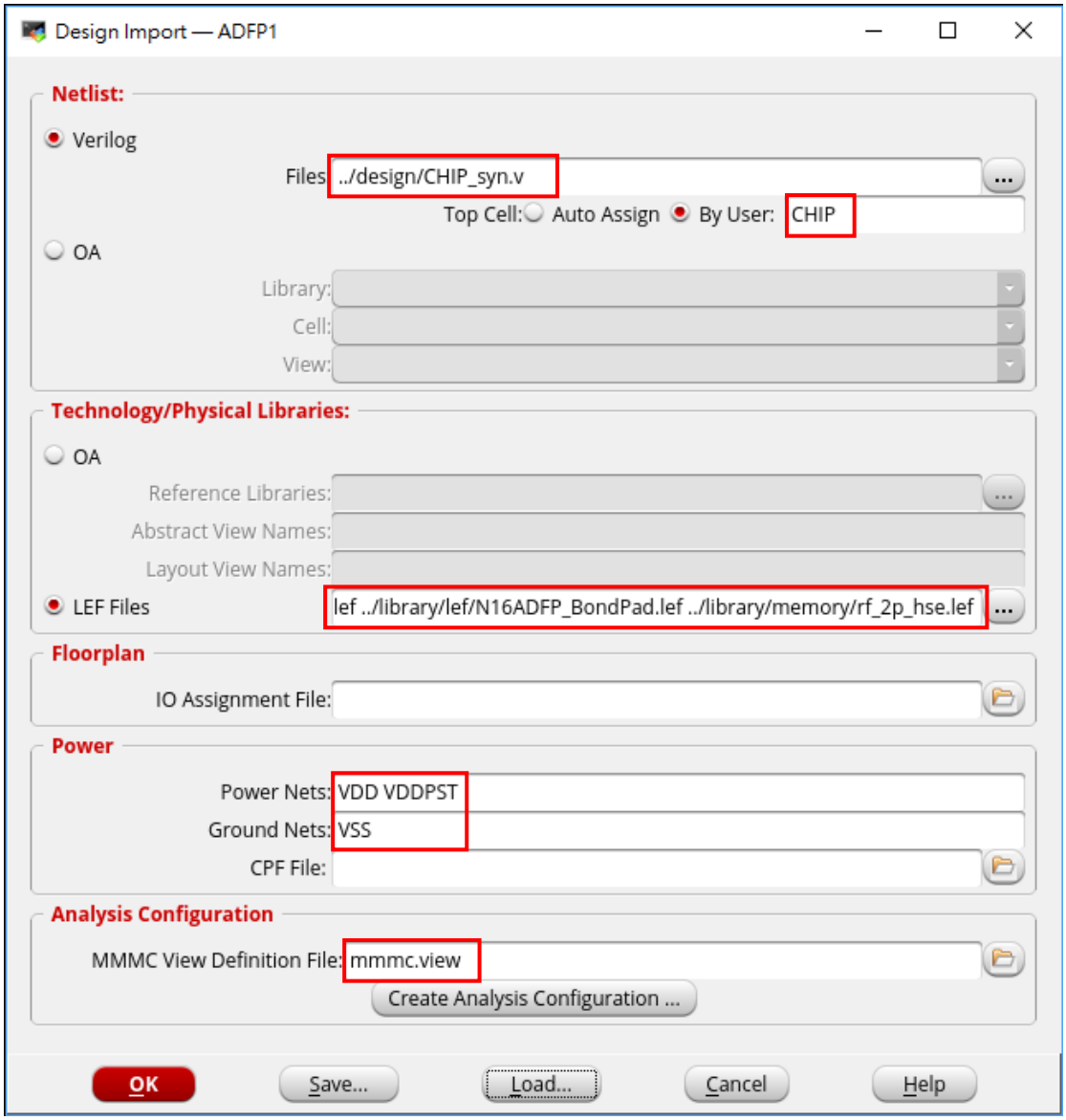
- Power Nets: **VDD VDDPST**
- Ground Nets: **VSS**

3.6 Multi-Mode-Multi-Corner

- MMMC View Definition File: **mmmc.view**

3.7 Save current settings:

- Click Save… button
- File name: **CHIP.conf**
- Click OK button



Note: The mmmc.view is the same setting as the following figure.



Add PG pad and Corner pad

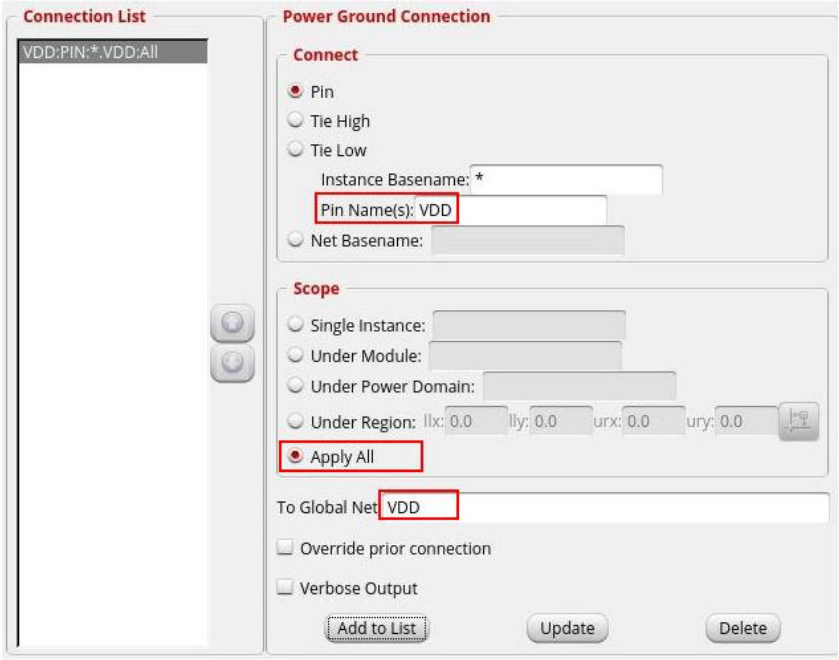
innovus # > **source lab_script/create_pg_pad.tcl**

```
if {[llength [get_db insts CORE_PG*]] == 0} {
    addInst -cell PVDD2CDGM_V -inst IO_PG1
    addInst -cell PVDD2CDGM_H -inst IO_PG2
    addInst -cell PVDD2CDGM_V -inst IO_PG3
    addInst -cell PVDD2CDGM_H -inst IO_PG4
    addInst -cell PVDD1CDGM_V -inst CORE_PG1
    addInst -cell PVDD1CDGM_H -inst CORE_PG2
    addInst -cell PVDD1CDGM_V -inst CORE_PG3
    addInst -cell PVDD1CDGM_H -inst CORE_PG4
}
if {[llength [get_db insts CORNER*]] == 0} {
    addInst -cell PCORNER -inst CORNERTL
    addInst -cell PCORNER -inst CORNERTR
    addInst -cell PCORNER -inst CORNERBL
    addInst -cell PCORNER -inst CORNERBR
}
```

Note: Can also add these pads in CHIP.io when doing import design.

Global Net Connect

1. Open ***Power → Connect Global Nets…***
2. Add all VDD pins to Connection List:
   - Connect ◆ Pin ◆ Pin Name(s): **VDD**
   - Scope ◆ **Apply All**
   - To Global Nets: **VDD**
   - Click Add to List button

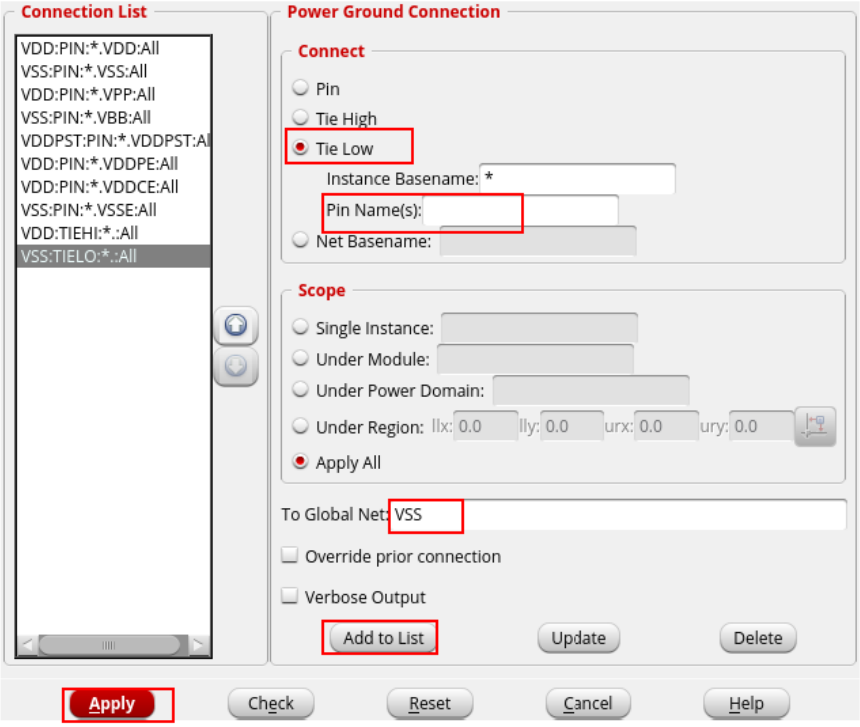


3. Add all VDDPE pins to Connection List:
   - Connect ◆ Pin ◆ Pin Name(s): **VDDPE**
   - Scope ◆ **Apply All**
   - To Global Nets: **VDD**
   - Click Add to List button
4. Add all VDDCE pins to Connection List:
   - Connect ◆ Pin ◆ Pin Name(s): **VDDCE**
   - Scope ◆ **Apply All**
   - To Global Nets: **VDD**
   - Click Add to List button
5. Add all VPP pins to Connection List:
   - Connect ◆ Pin ◆ Pin Name(s): **VPP**
   - Scope ◆ **Apply All**
   - To Global Nets: **VDD**
   - Click Add to List button
6. Add all VDDPST pins to Connection List:
   - Connect ◆ Pin ◆ Pin Name(s): **VDDPST**

VDDPE, VDDCE, VSSE: sram power pin name
VPP: standard cell pwell tap
VBB: standard cell nwell tap

- Scope ◆ **Apply All**
- To Global Nets: **VDDPST**
- Click Add to List button
-

7. Add all VSS pins to Connection List:
   - Connect ◆ Pin ◆ Pin Name(s): **VSS**
   - Scope ◆ **Apply All**
   - To Global Nets: **VSS**
   - Click Add to List button
8. Add all VSSE pins to Connection List:
   - Connect ◆ Pin ◆ Pin Name(s): **VSSE**
   - Scope ◆ **Apply All**
   - To Global Nets: **VSS**
   - Click Add to List button
9. Add all VBB pins to Connection List:
   - Connect ◆ Pin ◆ Pin Name(s): **VBB**
   - Scope ◆ **Apply All**
   - To Global Nets: **VSS**
   - Click Add to List button
10. Add all Tie High pins to Connection List:
   - Connect ◆ **Tie High**
   - Scope ◆ **Apply All**
   - To Global Nets: **VDD**
   - Click Add to List button
11. Add all Tie Low pins to Connection List:
   - Connect ◆ **Tie Low**
   - Scope ◆ **Apply All**
   - To Global Nets: **VSS**
   - Click Add to List button



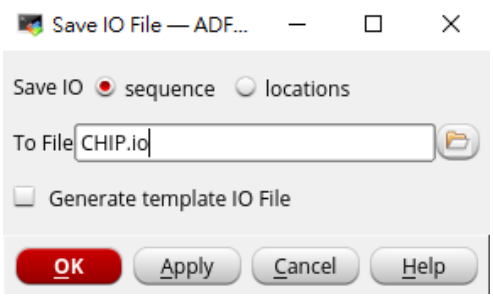
12. Apply the connection list and check:
    - Click Apply button
    - Click Check button
    - Click Cancel button (no message should be occured)

Specifying Scan Chain

1. innovus # > **defIn design/CHIP-scan.def**
2. innovus # > **report_scan_chain**

Define IO Constraint

1. Save IO file
    - Open ***File → Save → I/O File…***
    - Choose ◆ sequence
    - To File: **CHIP.io**
    - Click OK button



Note: Need to edit CHIP.io manually to organize the signal and power pads' orientation and order.

2. Load our provided IO constraint file
   - Open ***File → Load → I/O File…***
   - File name: **design/CHIP.io**
   - Click Open button

Set Configuration

1. Set global configuration
   - innovus # > **source lab_script/config.tcl**
2. Set clock tree synthesis (CTS) configuration
   - innovus # > **source lab_script/config_cts.tcl**

Note: These configurations may not be used immediately. But Set up these configurations in advance can ensure consistency throughout APR.

3. Save file
   - Open ***File → Save Design…***
   - Choose ◆ Innovus
   - File Name: **DBS/init.enc**
   - Click OK button

Floorplan

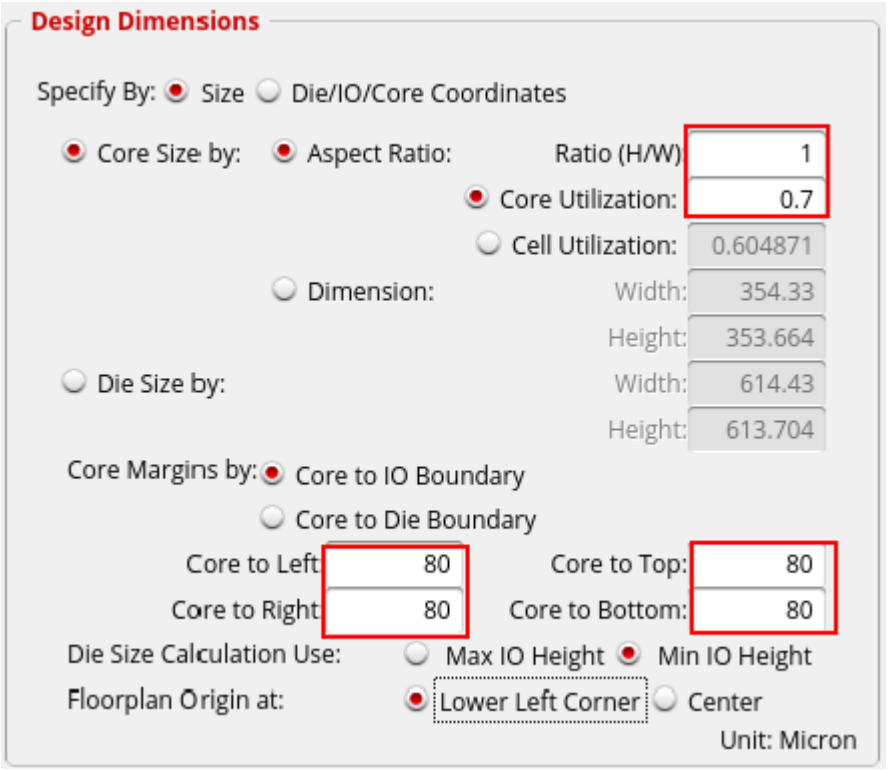
1. Open ***Floorplan → Specify Floorplan…***
2. Specify core size:
   - Ratio (H/W): 1
   - Core Utilization: 0.7
3. Specify core margin:
   - Core to IO Boundary

     Core to Left: **80**

     Core to Right: **80**

     Core to Top: **80**

     Core to Bottom: **80**
4. Click OK button

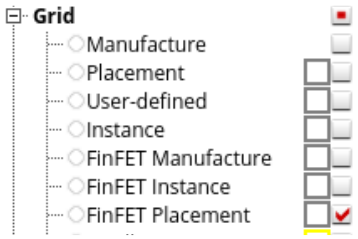


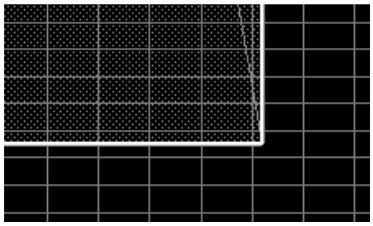
5. Load IO constraint file again
   - Open ***File → Load → I/O File…***
   - File name: **design/CHIP.io**
   - Click Open button
6. Update pad direction
   - Click pads and press "q" to observe the cells of pads
   - innovus # > **source lab_script/swap_io_hv.tcl**
   - innovus # > **do_swap_io**
   - Click pads and press "q" again to observe the cells of pads

Note: The left-hand and right-hand sides of the chip must place the horizontal pads (xxx_H). The top and bottom sides of the chip must place the vertical pads (xxx_V).

Fit FinFET Grid

1. Everything should be placed on grid to meet DRC
   - innovus # > **checkFPlan**
2. Open FinFET placement grid. Zoom in to the pad corner with DRC error. Can observe that the pad is not on grid.

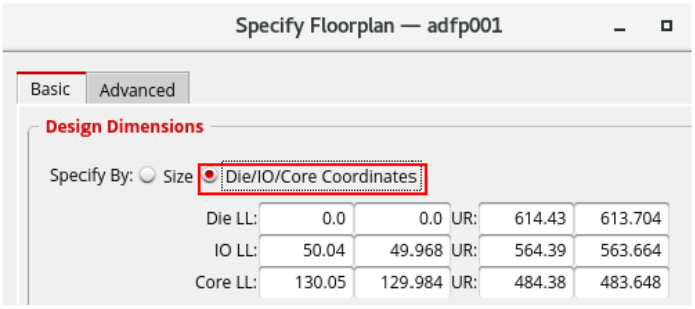




3. Fix off grid error
   - innovus # > **snapFPlan -all**
   - innovus # > **checkFPlan**

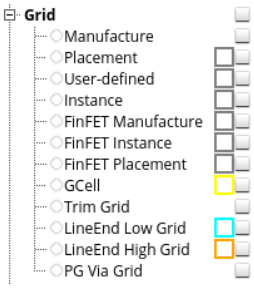
Note: Still has off grid error

4. Open ***Floorplan → Specify Floorplan…***
   - Choose ◆ Die/IO/Core Coordinates
   - Click OK button



   - innovus # > **loadIoFile design/CHIP.io -noAdjustDieSize**
   - innovus # > **snapFPlan -all**
   - innovus # > **checkFPlan**

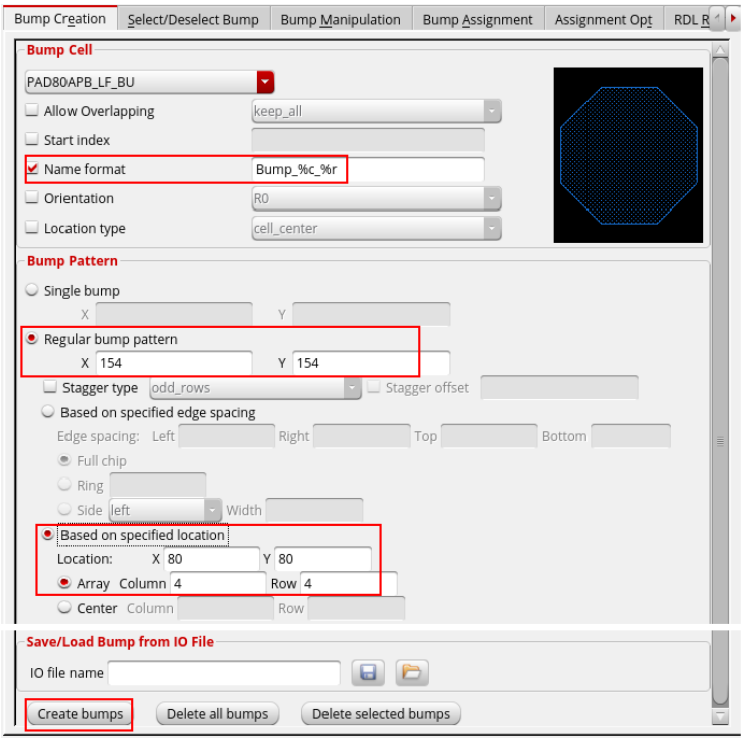
5. Close FinFET Placement display



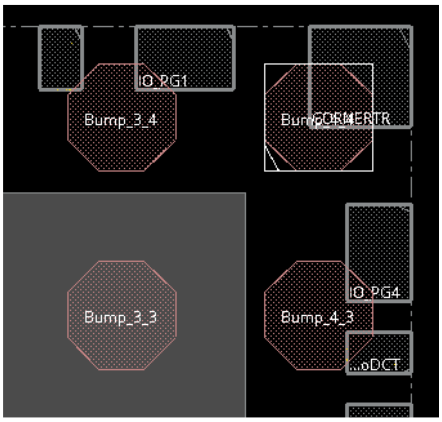
6. Save file
   - Open ***File → Save Design…***
   - Choose ◆ Innovus
   - File Name: **DBS/floorplan.enc**
   - Click OK button

Preroute Flip Chip Bump

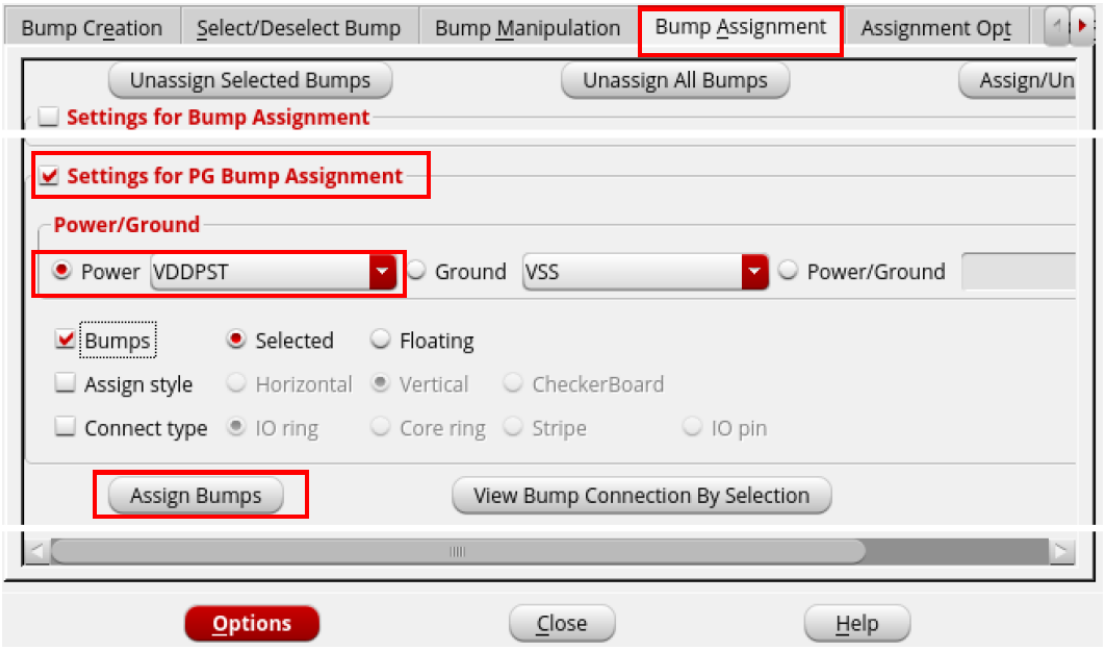
1. Check flip chip can be routed under this CHIP.io
   - Open ***Tools*** → ***Flip Chip…***



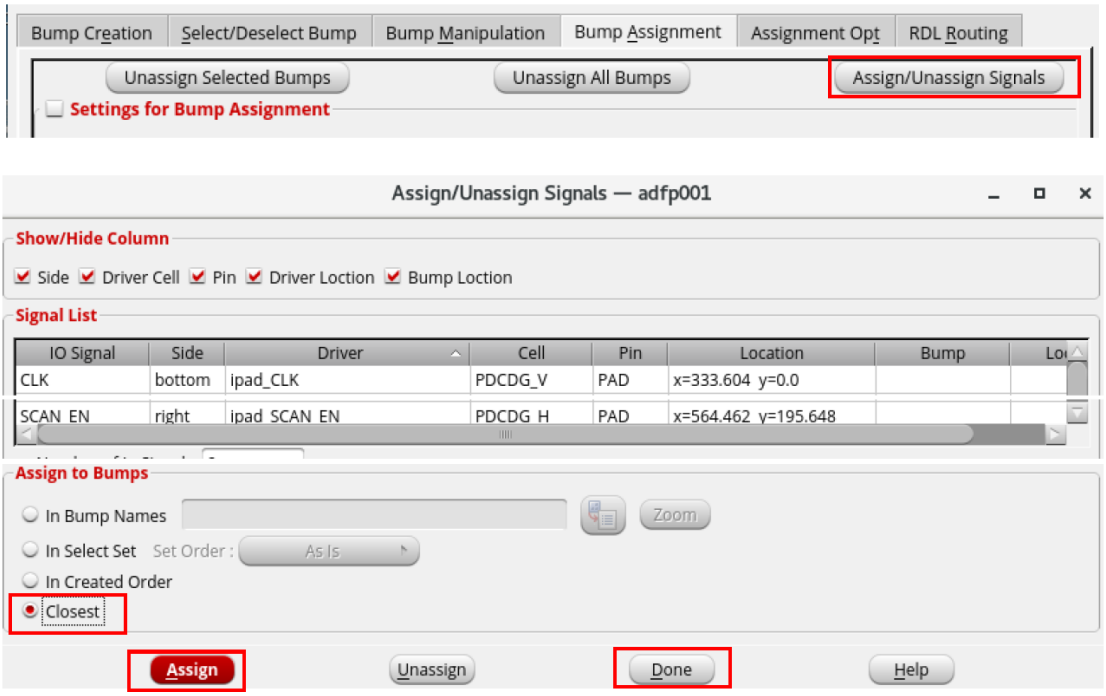
2. Assign PG bump
   - Click **Bump_4_4**



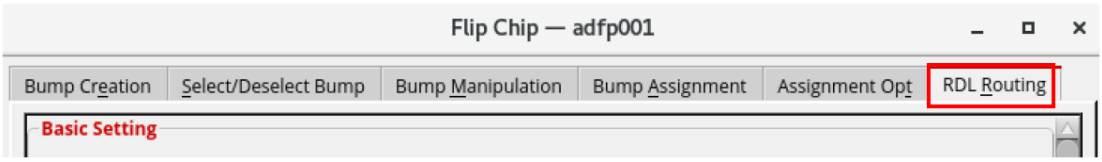
   - Choose ***Bump Assignment*** in the window
   - Set as the following figures. You should see **Bump_4_4** is assigned to VDDPST after this step.



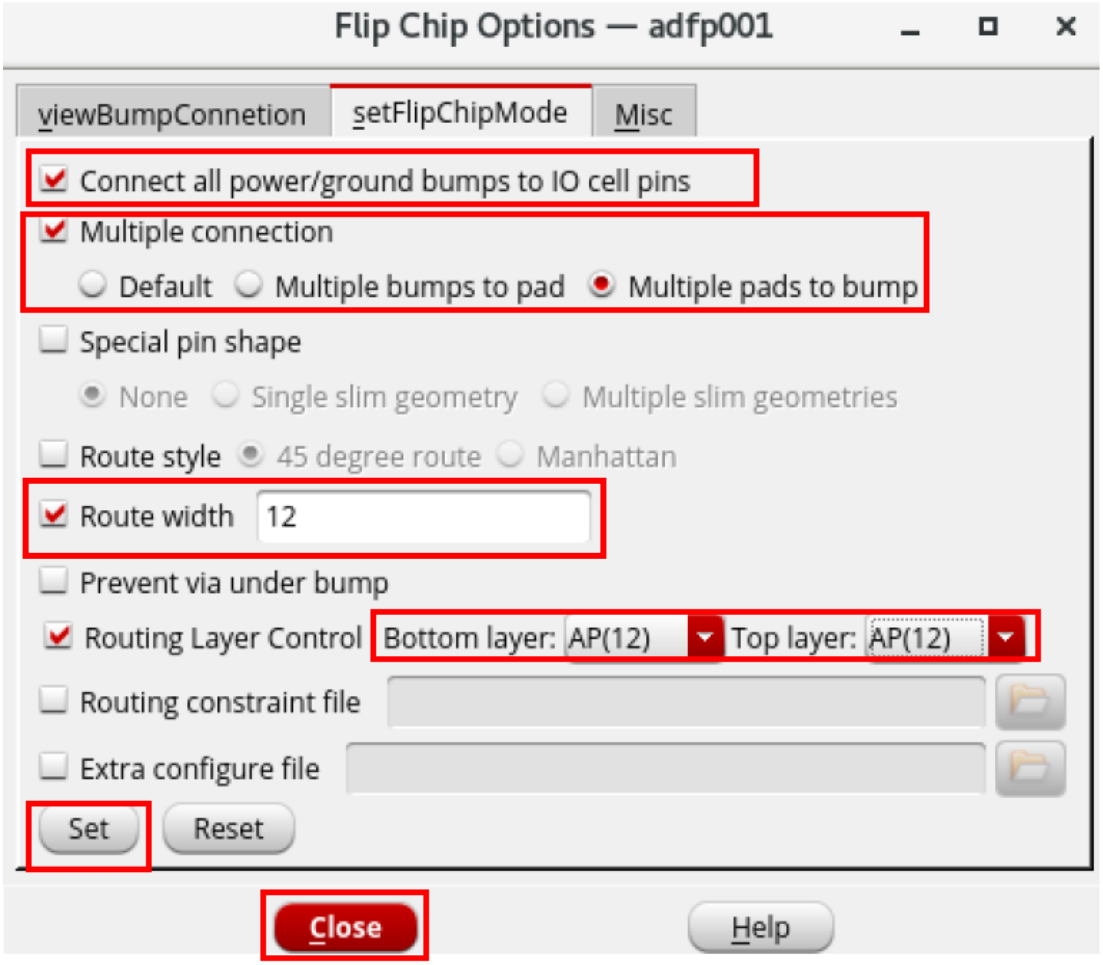
3. Observe the setting in lab_script/assign_pg_bumps.tcl and assign the other PG bumps.
   - innovus # > **source lab_script/assign_pg_bumps.tcl**
4. Click ***Assign/Unassign Signals***
   - Choose ◆ Closest
   - Click Assign button
   - Click Done button



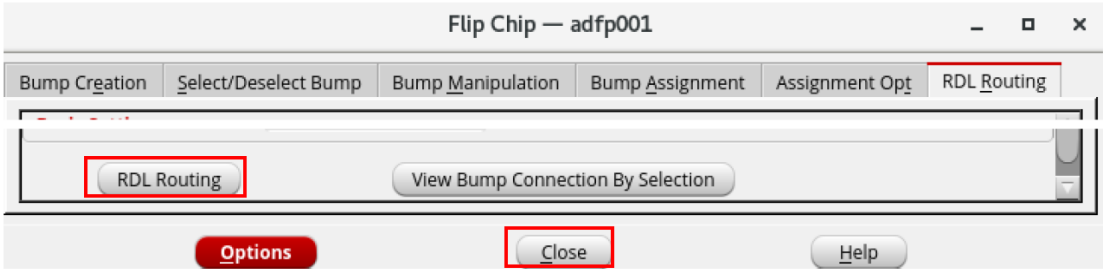
5. Choose ***RDL Routing*** in the window



- Click Options button
- Set as the following figures.



6. Back to the window
   - Click RDL Routing button
   - Click Close button



7. Delete bumps and RDL routing since we just want to make sure RDL route can be finished.

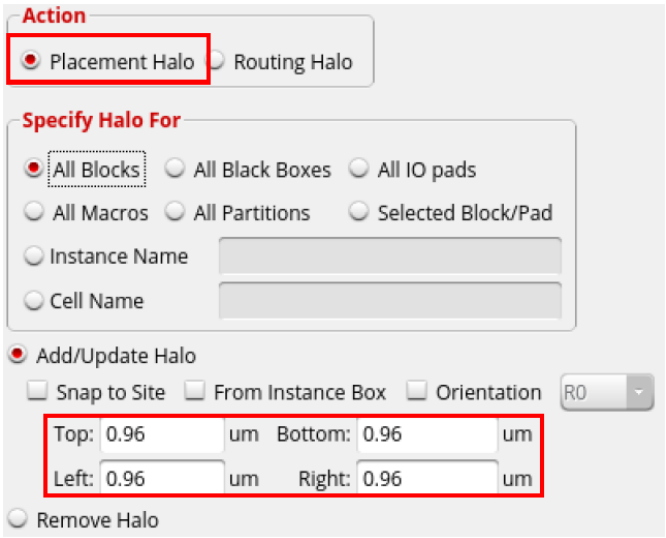
- innovus # > **editDelete -shapes iowire**
- innovus # > **deleteBumps -all**

Add IO Filler

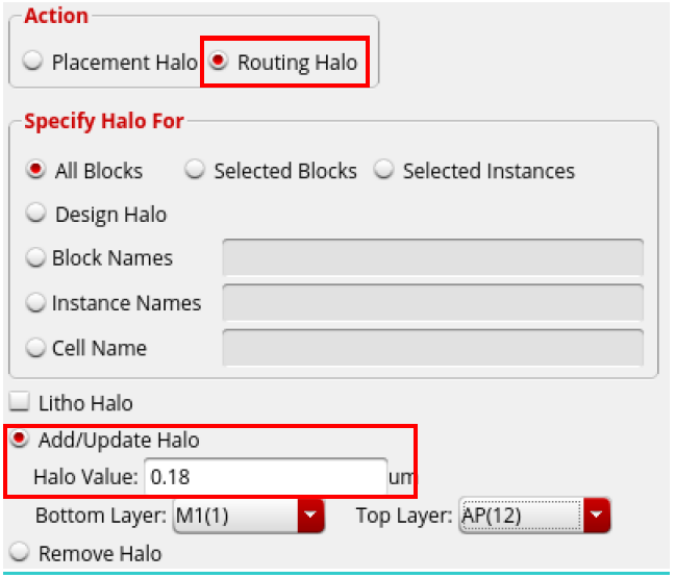
1. Fill the empty space between IO pads with IO fillers
   - innovus # > **source lab_script/add_io_fillers.tcl**
2. Fix IO pads and IO fillers
   - innovus # > **set_db [get_db insts -if {.base_cell.base_class == pad}] .place_status fixed**

Add Block Halo

1. Open ***Floorplan → Edit Floorplan → Edit Halo…***
   - Choose ◆ **Placement Halo**
   - Top/Left/Buttom/Right: **0.96 um**
   - Click Apply button



   - Choose ◆ **Routing Halo**
   - Halo Value: **0.18 um**
   - Click OK button

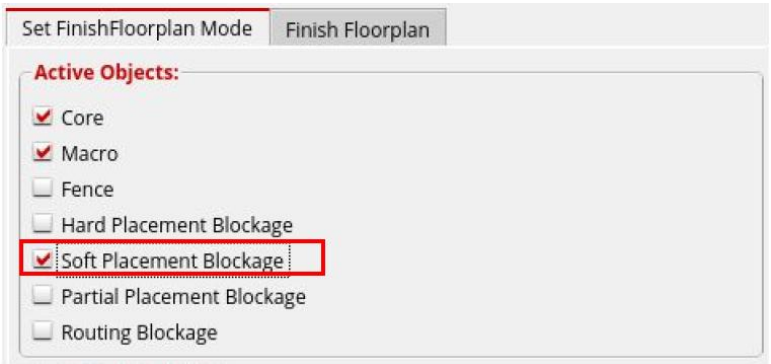


Place Macro

1. Set macro placement constraint
   - ➢ innovus # > **source lab_script/set_macro_place_constraint.tcl**
2. Place macro
   - ➢ innovus # > **place_design -concurrent_macros**
3. Check the macro placement meet the constraints or not. The place macro commands are not designed well. If not meet the constraints, move macros via Floorplan Toolbox (***Floorplan → Floorplan Toolbox…***).

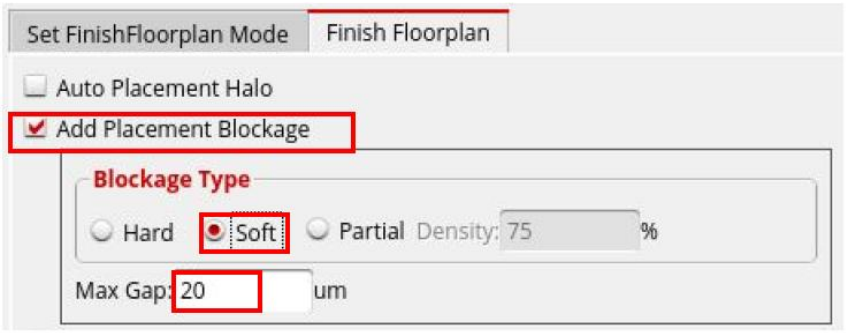
Note: Suggest that place all macros via Floorplan Toolbox.

4. Fix hard macro
   - ➢ innovus # > **setInstancePlacementStatus -allHardMacros -status fixed**
5. Open ***Floorplan → Automatic Floorplan → Finish Floorplan***
   - ➢ Choose ◆ **Soft Placement Blockage**



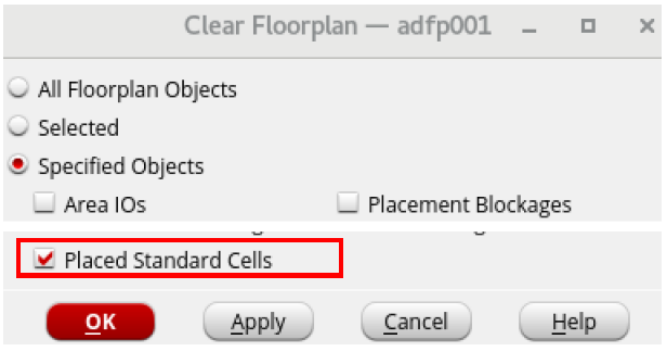
   - ➢ Choose ***Finish Floorplan*** in the window
   - ➢ Choose ◆ **Add Placement Blockage**
   - ➢ Choose ◆ **Soft**

- Max Gap: **20 um**
- Click OK button



Note: Add soft blockage in space <20 um.

6. Change to **placement view**  . Can observe that the core cells were placed in the core incorrectly due to **place_design -concurrent_macros**. Clear these core cells to prevent affecting powerplan.
   - Open ***Floorplan → Clear Floorplan…***
   - Choose ◆ **Placed Standard Cells**
   - Click OK button



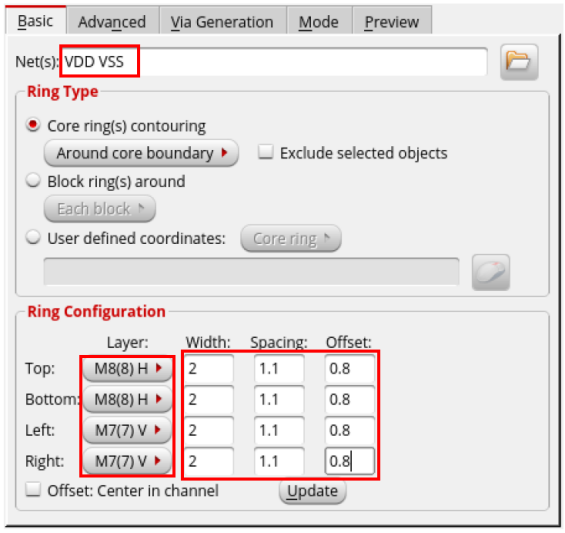
Note: If place all macros via Floorplan Toolbox, core cells would not be placed in the core.

7. Save file
   - Open ***File → Save Design…***
   - Choose ◆ Innovus
   - File Name: **DBS/floorplan.enc**

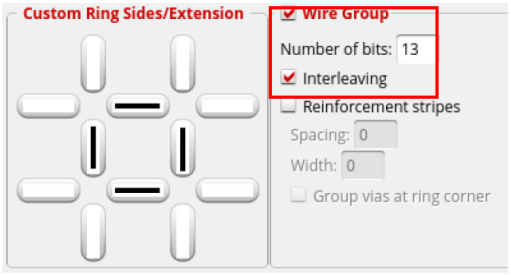
Power Planning: Power Ring

1. Create power ring. Add M8/M7 ring and M6/M5 ring.
   - Open ***Power → Power Planning → Add Ring…***
   - Nets(s): **VDD VSS**
   - Top: Layer: **M8(8) H** Width: **2** Spacing: **1.1** Offset: **0.8**
   - Bottom: Layer: **M8(8) H** Width: **2** Spacing: **1.1** Offset: **0.8**

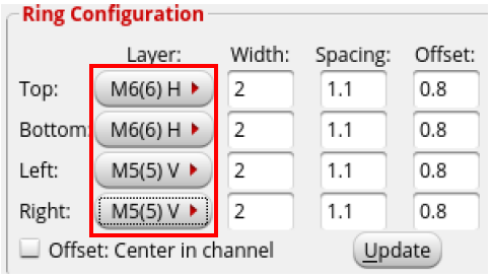
- Left: Layer: **M7(7) V** Width: **2** Spacing: **1.1** Offset: **0.8**
- Right: Layer: **M7(7) V** Width: **2** Spacing: **1.1** Offset: **0.8**



- Choose ***Advanced*** in the window
- Choose ◆ **Wire Group**
- Number of bits: 13
- Choose ◆ **Interleaving**
- Click Apply button



- Choose ***Basic*** in the window
- Modify Top/Bottom Layer to **M6(6) H** and Left/Right Layer to **M5(5) V**
- Click OK button



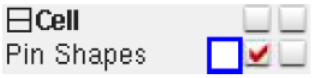
- innovus # > **editTrim -nets {VDD VSS} -layers {M8 M7 M6 M5}**

2. Save file
   - Open ***File → Save Design…***
   - Choose ◆ Innovus

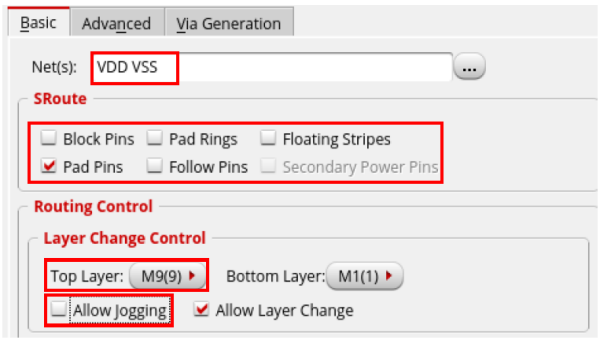
- File Name: **DBS/powerplan.enc**

Power Planning: Power Pad

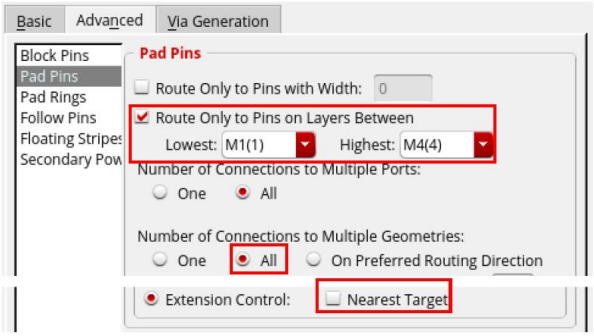
1. Set **Cell/Pin Shapes** to visible for observing the power pin connection



2. Open ***Route → Special Route…***
   - Net(s): **VDD VSS**
   - Preserve only ◆ **Pad Pins**
   - Top Layer: **M9(9)**
   - Unchosen **Allow jogging**



   - Choose ***Advanced*** in the window and choose ***Pad Pins***
   - Choose ◆ **Route Only to Pins on Layers Between**
   - Lowest: **M1(1)** Highest: **M4(4)**
   - Number of Connections to Multiple Geometries: ◆ **All**
   - Unchosen **Nearest Target**



   - Choose ***Via Generation*** in the window
   - Choose ◆ **Core Ring**
   - Click OK button



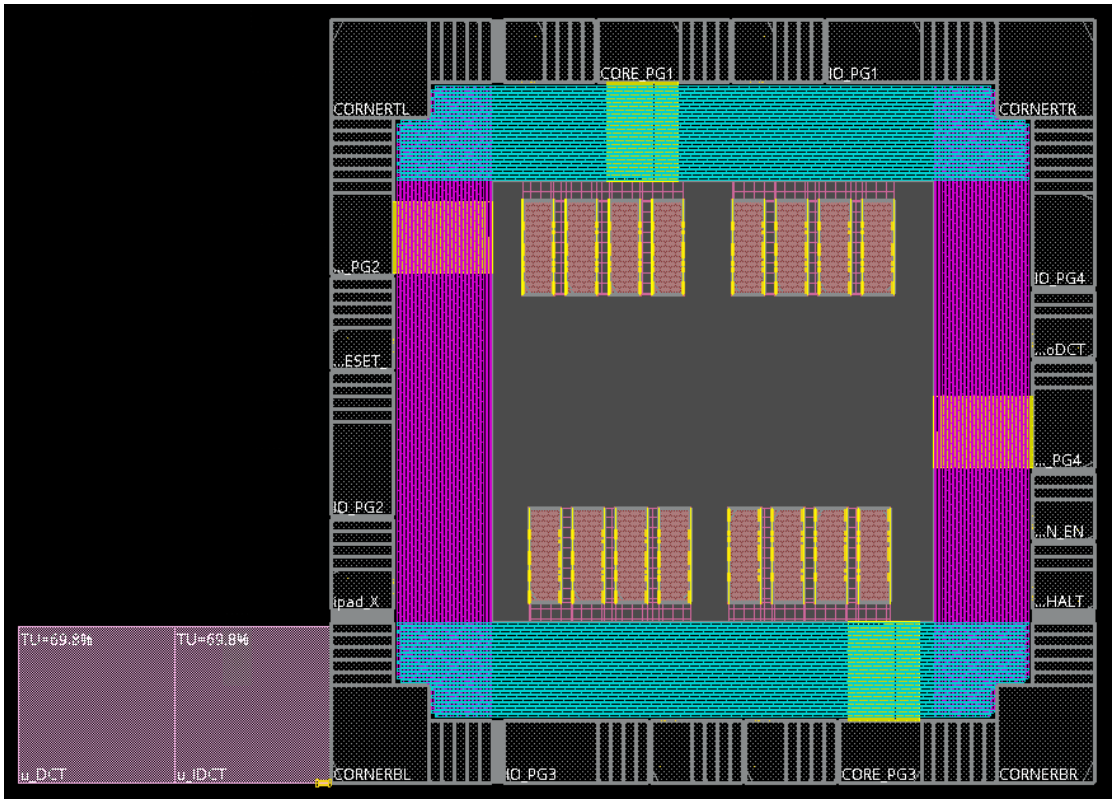
3. Add routing blockage between power pad and power ring to prevent stripe and followpin to route in these area for reducing DRC.

- innovus # > **source lab_script/create_padpin_blockage.tcl**

4. Add 0.46 um M1/M2/M3 routing blockage around the macros to prevent power route in these areas. Easier to route macro signal.
   - innovus # > **source lab_script/create_block_blockage.tcl**
5. Save file
   - Open ***File → Save Design…***
   - Choose ◆ Innovus
   - File Name: **DBS/powerplan.enc**

Checkpoints

1. Take a snapshot after floorplan.



2. **Due Tuesday, Apr. 21, 20:20**
3. Submit to NTU COOL in pdf format.

Appendix 1: Generate CHIP.ioc

1. Open ***File → Save → I/O File…***
   - Save IO ◆ sequence
   - To File: **CHIP.ioc**
   - ◆ Generate template IO File
   - Click OK button

Do after Import Design

2. Open ***File → Load → I/O File…***

- Choose **CHIP.ioc**
- Click Open button